Dynamic Semantics of ReOCL

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This is a companion document for the "ReOCL: An Incremental Core of OCL for Reactive Web-Based Modeling" paper.

Contents

1 Value Domain

$$v \in \mathsf{Val} ::= \mathsf{VTrue} \mid \mathsf{VFalse} \mid \mathsf{VInt}(n) \mid \mathsf{VString}(s) \mid \mathsf{VObj}(oid, C) \mid \mathsf{VColl}(vs)$$

$$[\![e]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) \in \mathsf{option}(\mathsf{Val}) ::= \mathsf{Some}(v) \mid \mathsf{None}$$

where γ is the variable environment, σ is the current store, and σ_{pre} is the pre-state snapshot mapping locations to optional values.

2 Auxiliary Functions

$$\mathsf{boolVal}(b) \triangleq \begin{cases} \mathsf{VTrue}, & \text{if } b = \mathsf{true} \\ \mathsf{VFalse}, & \text{otherwise} \end{cases}$$

$$\mathsf{expectBool}(v) \triangleq \begin{cases} \mathsf{Some}(\mathsf{true}), & \text{if } v = \mathsf{VTrue} \\ \mathsf{Some}(\mathsf{false}), & \text{if } v = \mathsf{VFalse} \\ \mathsf{None}, & \text{otherwise} \end{cases}$$

$$\mathsf{expectInt}(v) \triangleq \begin{cases} \mathsf{Some}(n), & \text{if } v = \mathsf{VInt}(n) \\ \mathsf{None}, & \text{otherwise} \end{cases}$$

$$\mathsf{expectObj}(v) \triangleq \begin{cases} \mathsf{Some}(oid, C), & \text{if } v = \mathsf{VObj}(oid, C) \\ \mathsf{None}, & \text{otherwise} \end{cases}$$

$$\mathsf{expectColl}(v) \triangleq \begin{cases} \mathsf{Some}(vs), & \text{if } v = \mathsf{VColl}(vs) \\ \mathsf{None}, & \text{otherwise} \end{cases}$$

$$\mathsf{readField}(\sigma, oid, C, f) \triangleq \mathsf{Some}(\sigma(\mathsf{fieldStateId}(C, oid, f)))$$

$$\mathsf{readPre}(\sigma_{\mathsf{pre}}, oid, C, f) \triangleq \sigma_{\mathsf{pre}}(\mathsf{fieldStateId}(C, oid, f))$$

3 Evaluation Rules

3.1 Literals and Variables

$$\frac{}{[\![\mathsf{true}]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \mathsf{Some}(\mathsf{VTrue})}\ \textsc{E-True}$$

$$\frac{}{[\![\mathsf{false}]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \mathsf{Some}(\mathsf{VFalse})}\ \textsc{E-False}$$

$$\frac{}{[\![n]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \mathsf{Some}(\mathsf{VInt}(n))}\ \textsc{E-IntLit}$$

$$\frac{}{[\![s]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \mathsf{Some}(\mathsf{VString}(s))}\ \textsc{E-StringLit} \qquad \frac{}{[\![\mathsf{self}]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \gamma(\mathsf{self})}\ \textsc{E-Self}$$

$$\frac{}{[\![x]\!](\gamma, \sigma, \sigma_{\mathsf{pre}}) = \gamma(x)}\ \textsc{E-Var}$$

3.2 Navigation

$$\frac{[\![e]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C))}{[\![e.f]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{readField}(\sigma, oid, C, f)}\ \text{E-Nav}$$

$$\frac{[\![e]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C))}{[\![e.f@\text{pre}]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{readPre}(\sigma_{\text{pre}}, oid, C, f)}\ \text{E-Pre}$$

3.3 Boolean Connectives

$$\frac{[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}{[\![e_1 \text{ and } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}\ \text{E-And-False}$$

$$\frac{\begin{array}{c}[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VTrue}) \\ [\![e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v) \quad \text{expectBool}(v) = \text{Some}(b)\end{array}}{[\![e_1 \text{ and } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(b))}\ \text{E-And-True}$$

$$\frac{[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}{[\![e_1 \text{ or } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-Or-True}$$

$$\frac{\begin{array}{c}[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VFalse}) \\ [\![e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v) \quad \text{expectBool}(v) = \text{Some}(b)\end{array}}{[\![e_1 \text{ or } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(b))}\ \text{E-Or-False}$$

$$\frac{[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}{[\![e_1 \text{ implies } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-Implies-False}$$

$$\frac{\begin{array}{c}[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{VTrue}) \\ [\![e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v) \quad \text{expectBool}(v) = \text{Some}(b)\end{array}}{[\![e_1 \text{ implies } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(b))}\ \text{E-Implies-True}$$

$$\frac{\begin{array}{c}[\![e_1]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v_1) \quad [\![e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v_2) \\ \text{expectBool}(v_1) = \text{Some}(b_1) \quad \text{expectBool}(v_2) = \text{Some}(b_2)\end{array}}{[\![e_1 \text{ xor } e_2]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(b_1 \oplus b_2))}\ \text{E-Xor}$$

$$\frac{[\![e]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(v) \quad \text{expectBool}(v) = \text{Some}(b)}{[\![\text{not } e]\!](\gamma, \sigma, \sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(\neg b))}\ \text{E-Not}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue}) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v)}{[\![\text{if } e_1 \text{ then } e_2 \text{ else } e_3 \text{ endif}]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v)}\ \text{E-If-True}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse}) \quad [\![e_3]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v)}{[\![\text{if } e_1 \text{ then } e_2 \text{ else } e_3 \text{ endif}]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v)}\ \text{E-If-False}$$

3.4 Equality

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v_1) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v_2)}{[\![e_1 = e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(v_1 = v_2))}\ \text{E-Eq}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v_1) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(v_2)}{[\![e_1 <> e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(\neg(v_1 = v_2)))}\ \text{E-Neq}$$

Equality is structural: $\text{VTrue} \neq \text{VInt}(0)$, $\text{VColl}(vs_1) = \text{VColl}(vs_2)$ iff element-wise equality.

3.5 Arithmetic

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 + e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n+m))}\ \text{E-Add}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 - e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n-m))}\ \text{E-Sub}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 * e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n \cdot m))}\ \text{E-Mul}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m)) \quad m \neq 0}{[\![e_1 / e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n \div m))}\ \text{E-Div}$$

Division by zero yields None (no rule applies).

3.6 Comparisons

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 < e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(n < m))}\ \text{E-Lt}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 > e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(n > m))}\ \text{E-Gt}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 <= e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(n \leq m))}\ \text{E-Leq}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(n)) \quad [\![e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(m))}{[\![e_1 >= e_2]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(n \geq m))}\ \text{E-Geq}$$

3.7 View-Producing Operators

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(r_v),\quad r_v \in \{\text{VTrue},\text{VFalse}\}}{[\![e_1 \to \text{select}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(\{v \in vs \mid r_v = \text{VTrue}\}))}\ \text{E-Select}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(r_v),\quad r_v \in \{\text{VTrue},\text{VFalse}\}}{[\![e_1 \to \text{reject}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(\{v \in vs \mid r_v = \text{VFalse}\}))}\ \text{E-Reject}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(w_v)}{[\![e_1 \to \text{collect}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}([w_v \mid v \in vs]))}\ \text{E-Collect}$$

3.8 Terminal Operators

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}{[\![e_1 \to \text{forAll}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-ForAll-True}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \exists v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}{[\![e_1 \to \text{forAll}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}\ \text{E-ForAll-False}$$

ForAll may short-circuit once a false witness is found. Undefined predicate evaluations that are not needed to determine a false result are not observed.

Exists may short-circuit once a true witness is found. Undefined predicate evaluations that are not needed to determine a true result are not observed.

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \exists v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}{[\![e_1 \to \text{exists}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-Exists-True}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}{[\![e_1 \to \text{exists}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}\ \text{E-Exists-False}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(r_v),\quad r_v \in \{\text{VTrue},\text{VFalse}\} \quad |\{v \in vs \mid r_v = \text{VTrue}\}| = 1}{[\![e_1 \to \text{one}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-One}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(r_v),\quad r_v \in \{\text{VTrue},\text{VFalse}\} \quad |\{v \in vs \mid r_v = \text{VTrue}\}| \neq 1}{[\![e_1 \to \text{one}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}\ \text{E-One-False}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(w_v) \quad \neg\exists i \neq j.\ w_i = w_j}{[\![e_1 \to \text{isUnique}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})}\ \text{E-IsUnique}$$

$$\frac{[\![e_1]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ [\![e_2]\!](\gamma[x \mapsto v],\sigma,\sigma_{\text{pre}}) = \text{Some}(w_v) \quad \exists i \neq j.\ w_i = w_j}{[\![e_1 \to \text{isUnique}(x \mid e_2)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})}\ \text{E-IsUnique-False}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs))}{[\![e \to \text{size}()]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(|vs|))}\ \text{E-Size}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs))}{[\![e \to \text{isEmpty}()]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(vs = []))}\ \text{E-IsEmpty}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs))}{[\![e \to \text{notEmpty}()]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{boolVal}(vs \neq []))}\ \text{E-NotEmpty}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VColl}(vs)) \quad \forall v \in vs.\ v = \text{VInt}(n_v)}{[\![e \to \text{sum}()]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VInt}(\textstyle\sum_{v \in vs} n_v))}\ \text{E-Sum}$$

Sum requires all elements to be integers; otherwise None.

3.9 Type Tests

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C')) \quad C' \sqsubseteq C}{[\![e \to \text{oclIsKindOf}(C)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})} \text{ E-KindOf-True}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C')) \quad C' \not\sqsubseteq C}{[\![e \to \text{oclIsKindOf}(C)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})} \text{ E-KindOf-False}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C')) \quad C' = C}{[\![e \to \text{oclIsTypeOf}(C)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VTrue})} \text{ E-TypeOf-True}$$

$$\frac{[\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VObj}(oid, C')) \quad C' \neq C}{[\![e \to \text{oclIsTypeOf}(C)]\!](\gamma,\sigma,\sigma_{\text{pre}}) = \text{Some}(\text{VFalse})} \text{ E-TypeOf-False}$$

Both type tests require an object receiver; on non-object values they yield None.

3.10 Invariant Validity

$$\text{isTrue}(r) \;\Leftrightarrow\; r = \text{Some}(\text{VTrue})$$

$$\text{validInv}(e,\gamma,\sigma,\sigma_{\text{pre}}) \;\Longleftrightarrow\; \text{isTrue}([\![e]\!](\gamma,\sigma,\sigma_{\text{pre}}))$$

$$\text{evalInvariant}(selfOid,\sigma,\sigma_{\text{pre}},inv) = [\![\text{invBody}(inv)]\!](ve,\sigma,\sigma_{\text{pre}}),$$

where $ve = [(\text{self}, \text{VObj}(selfOid, \text{invContext}(inv)))]$.